

# Arabizi-Arabic Neural Transliteration System Final Model: v14 Pipeline with v12-Base Transformer

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# Problem



## The Arabizi Problem

- Arabic script is not compatible with standard keyboards
- Social media users use Roman letters (Arabizi)
- Romanization depends on local dialect and colonial language

## Intelligence & Security Implications

- OSINTers struggle to search Arabic social media
- Terrorist/hate speech monitoring misses Arabizi content
- LE cannot query Arabic databases with Romanized input

## Current Tools Gap

- Existing transliteration tools are rule-based and inaccurate
- No dialect aware bidirectional neural approaches conversion tools
- Need for real-time, accurate transliteration at scale

# Solution

## Arabizi-Arabic Transliterator

A Transformer-based neural transliteration system

Bidirectional conversion: Arabizi ↔ Arabic

Real-time web/API demo with top-k outputs

## Key Features

Separate sequence-to-sequence models per direction

Beam search decoding for stronger candidate recall

Learned MLP re-ranker after beam generation

Unicode normalization, diacritic handling, and Arabizi digit tokens

Returns top-12 predictions with beam/reranker scores

The screenshot shows a web application titled "Arabizi ↔ Arabic Translator" with a logo. It features two dropdown menus at the top: "Arabic → Arabizi" and "Palestinian Arabic". The main interface is split into two panels. The left panel, labeled "Arabic" with the instruction "Type Arabic text", contains the input text "سلام". The right panel, labeled "Arabizi" with the instruction "All returned transliterations · 0:08", displays a list of six transliteration candidates: 1. salaam, 2. salam, 3. saalim, 4. silmaam, 5. s2aalam, and 6. sal2aam. Below the input panels are three buttons: "Clear", "Copy Results", and "Transliterate". At the bottom, a "Dialect options" section shows a grid of buttons for different dialects: "Palestinian Arabic" (marked "Available"), "Levantine Arabic" (marked "Coming later"), "Egyptian Arabic" (marked "Coming later"), "Gulf Arabic" (marked "Coming later"), "Iraqi Arabic" (marked "Coming later"), and "Maghrebi Arabic" (marked "Coming later").

# High-Level Architecture

## Architecture Components

Transformer encoder-decoder models

Encoder: 4 layers, 8 attention heads

Decoder: 4 layers, 8 attention heads

Embedding dimension: 256

Feed-forward dimension: 1024

Dropout: 15%

## Input Processing

Direction-specific tokenization

Arabic: character-level tokens

Arabizi: character + multi-token patterns like sh, kh, 3a, 7o

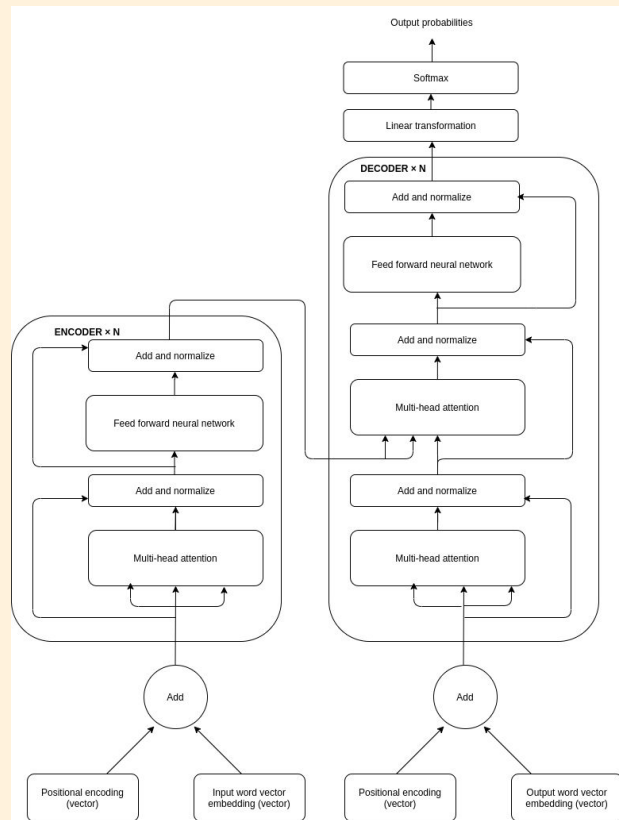
Special tokens: <PAD>, <BOS>, <EOS>, <UNK>

Unicode normalization (NFC) + lowercase Arabizi

## Output Generation

Beam search width=20 with length penalty

Returns top-12 predictions per input



# Architecture Deep Dive

## Vocabulary

Direction-specific source/target vocabularies

Arabic vocab: 44 tokens

Arabizi vocab: 54–65 tokens depending on direction

Covers Arabic letters/diacritics, Roman letters, digits, spaces, and common digraphs

## Training Configuration

Loss: cross-entropy with 0.02 label smoothing

Optimizer: AdamW with cosine learning-rate schedule

Batch size: 96; max 100 epochs; early stopping patience 16

Length curriculum + capped weighted sampling

Checkpoints and metrics autosaved to Google Drive

## Deployment Artifacts

Base model website bundles include tokenizer, weights, vocab, and config

Re-ranker .pt files are saved separately and loaded with the backend

# Result Re-Ranker

## Why Re-Ranking

Arabizi is highly ambiguous: one Romanized input can map to multiple Arabic forms

Beam search often contains the right answer, but not always at rank 1

Re-ranking uses extra candidate signals beyond raw sequence likelihood

## Re-Ranking Model

Lightweight MLP trained separately for each direction

Uses beam candidates generated by the Transformer

Training capped at 5,000 source rows per direction for compute efficiency

Runs after beam search to reorder the top candidate list

## Candidate Features

Beam rank, score gap, raw and normalized beam scores

Source/candidate length features and ratios

Arabizi digit usage: 2, 3, 5, 6, 7, 8, 9

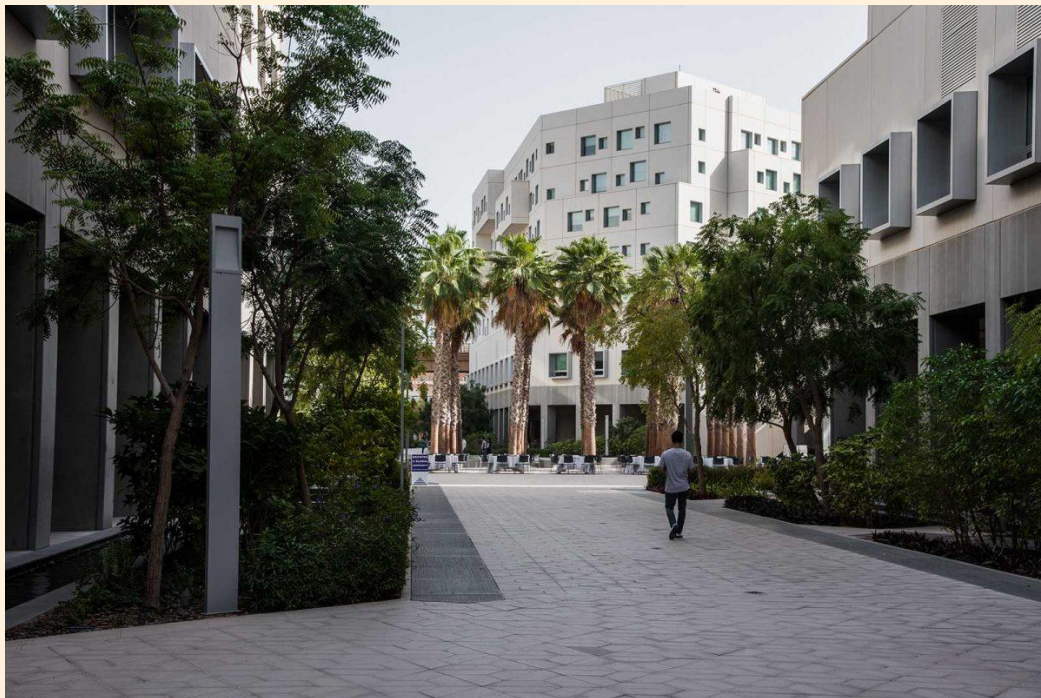
Space, vowel-like, and script-character patterns

Cached lexicon features: 50k Arabic words + 26,740 Arabizi forms

## Impact

Improves user-facing top-k alternatives while keeping inference lightweight

# Training and Data Preprocessing



## **Dataset: Maknuune v1.0.1**

From NYU Abu Dhabi

Parallel Palestinian Arabizi-Arabic pairs

34,816 cleaned paired rows after  
deduplication/cleaning

Pair split: 26,754 train / 3,344 validation / 3,345 test

## **Preprocessing Steps**

Unicode normalization (NFC form)

Arabic tatweel/kashida removal

Whitespace normalization

Text stripping and cleaning

Direction-specific tokenization

Arabizi-aware augmentation for training variants

## **Task Construction**

Arabic→Arabizi train rows: 26,740

Arabizi→Arabic train rows: 54,681 after  
augmentation

# Model Performance

## Metrics

Normalized exact match and practical exact match

Top-k beam hit rate and mean reciprocal rank (MRR)

Direction-specific evaluation: Arabizi→Arabic and Arabic→Arabizi

Full held-out test set: 6,686 direction-specific examples

## Final Test Results

Overall strict exact match: 69.91%

Overall practical exact match: 72.18%

Top-k strict exact match: 88.17%

Top-k practical exact match: 89.64%

Average edit distance: 1.28

## By Direction

Arabizi→Arabic: 72.36% strict; 76.91% practical; 90.88% top-k practical

Arabic→Arabizi: 67.45% strict; 88.39% top-k strict

## Beam Search & Re-Ranking

Beam top-3 strict hit rate: 82.23%

Beam top-5 strict hit rate: 85.13%

Beam top-10 strict hit rate: 87.65%

Beam MRR: 0.766

Arabizi → Arabic

Palestinian Arabic

Arabizi

Type Arabizi text

salaam

Clear

Copy Results

Transliterate

Dialect options

Palestinian  
Arabic

Available

Levantine  
Arabic

Coming  
later

Egyptian  
Arabic

Coming  
later

Gulf Arabic

Coming later

Iraqi Arabic

Coming later

Maghrebi  
Arabic

Coming  
later

Arabic

All returned transliterations · 0:08

1. سلام
2. صلام
3. لسام
4. ئلام
5. سلامة
6. سلامي



Arabic → Arabizi



Palestinian Arabic



Arabic

Type Arabic text

سلام

Clear

Copy Results

Transliterate

Dialect options

Palestinian  
Arabic

Available

Levantine  
Arabic

Coming  
later

Egyptian  
Arabic

Coming  
later

Gulf Arabic

Coming later

Iraqi Arabic

Coming later

Maghrebi  
Arabic

Coming  
later

Arabizi

All returned transliterations · 0:07

1. salaam
2. salam
3. saalim
4. silmaam
5. s2aalam
6. sal2aam



Arabizi → Arabic

Palestinian Arabic

Arabizi

Type Arabizi text

keef

Arabic

All returned transliterations · 0:07

1. كيف
2. كيفة
3. كافة

Clear

Copy Results

Transliterate

Dialect options

Palestinian  
Arabic

Available

Levantine  
Arabic

Coming  
later

Egyptian  
Arabic

Coming  
later

Gulf Arabic

Coming later

Iraqi Arabic

Coming later

Maghrebi  
Arabic

Coming  
later



Arabic → Arabizi



Palestinian Arabic



Arabic

Type Arabic text

كيف

Clear

Copy Results

Transliterate

Dialect options

Palestinian  
Arabic

Available

Levantine  
Arabic

Coming  
later

Egyptian  
Arabic

Coming  
later

Gulf Arabic

Coming later

Iraqi Arabic

Coming later

Maghrebi  
Arabic

Coming  
later

Arabizi

All returned transliterations · 0:07

1. kafi
2. kiif
3. fiik
4. keef
5. ykaf
6. kafa



Arabizi → Arabic

Palestinian Arabic

Arabizi

Type Arabizi text

qahwe

Arabic

All returned transliterations · 0:09

1. قهوة
2. قحوة
3. حقوة
4. قهوةٲة
5. قهوةزة

Clear

Copy Results

Transliterate

Dialect options

Palestinian  
Arabic

Available

Levantine  
Arabic

Coming  
later

Egyptian  
Arabic

Coming  
later

Gulf Arabic

Coming later

Iraqi Arabic

Coming later

Maghrebi  
Arabic

Coming  
later



Arabic → Arabizi



Palestinian Arabic



Arabic

Type Arabic text

قهوة

Clear

Copy Results

Transliterate

Dialect options

Palestinian  
Arabic

Available

Levantine  
Arabic

Coming  
later

Egyptian  
Arabic

Coming  
later

Gulf Arabic

Coming later

Iraqi Arabic

Coming later

Maghrebi  
Arabic

Coming  
later

Arabizi

All returned transliterations · 0:07

1. qahwe
2. qahwwe
3. hwuqkshe
4. waqahwe
5. hwuqkkshe
6. qahwekahwe

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[https://github.com/kagoeckner/arabizi\\_arabic\\_transliterator](https://github.com/kagoeckner/arabizi_arabic_transliterator)

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